Same patient, different SUV

Work in pairs. Where you see ✎ **Predict**, write your answer **before** you click. The task numbers match the tasks on the web pages. One patient throughout: a woman of 58 with lung cancer. **Order: C, D, break, B, A.**

patrikbrynolfsson.github.io/estro2026-dwi-handson/pet.html

Names _____ Specialty _____

C Reconstruction and lesion size

Start here · a simulated scan of the patient

1 · ITERATIONS AND FILTER

1 ✎ **Predict** More iterations: what happens to each?

Tumour SUVmax ☐ up ☐ same ☐ down Liver mean ☐ up ☐ same ☐ down

Noise ☐ up ☐ same ☐ down

2 Filter: None.

	1 iteration	3 iterations	12 iterations
Tumour SUVmax			
Liver noise (SD, % of mean)			

3 3 iterations + 5 mm filter: SUVmax _____ What the filter does: _____

4–5 The hospital's reconstruction: SUVmax 12.49. Her real SUVmax? _____

Follow-up in another hospital: what do you ask? _____

2 · LESION SIZE a phantom: six spheres, all with the same activity

1 ✎ **Predict** Will all six spheres show the same maximum? ☐ All the same ☐ Small ones lower ☐ Small ones higher

2–3 3 it + 5 mm, recovery: 10 mm _____ % 37 mm _____ % Read about right from _____ mm

4 All six inside the EARL range with: _____ Only one? ☐ Yes ☐ No

5 Bonus: a 1 cm node reads SUVmax 3. True SUV about _____

D Contouring on PET

2nd · same tumour · volumes in mL · then the break

1 · DRAW

1–3 Your outline: _____ mL Window ☐ 0-5 ☐ 0-10 ☐ 0-20

Another window: drawn differently? _____

2 · AUTOMATIC METHODS

1–3 ✎ **Predict** Tick the biggest (B) and the smallest (S) first. Then fill in the table.

	41 % of SUVmax	50 % of SUVmax	SUV 2.5	Adaptive	Gradient	Yours
Predict B / S						
Volume (mL)						
Overlap with expert						

4–5 SUV 2.5 at the tumour's top and bottom: what goes wrong? _____

One of these as the GTV? ☐ Yes ☐ No Instead: _____

3 · ANOTHER RECONSTRUCTION

1–2 ✎ **Predict** 12 iterations, no filter: the 41 % outline gets ☐ bigger ☐ the same ☐ smaller

Reconstruction	SUVmax	41 % volume	SUV 2.5 volume
1 iteration, 7 mm filter			
3 iterations, 5 mm filter			
12 iterations, no filter			

3–4 Why did the 41 % volume change? _____ Phantom: right for _____ mm

Patient: FDG-PET-CT-Lesions (autoPET), PETCT_7faf36a152 (Gatidis et al., Sci Data 2022;9:601; TCIA, doi:10.7937/gkr0-xv29; CC BY 4.0). Simulated scans (Blocks B-D) of the same patient: 4 mm pixels, scanner blur 6.5 mm, OSEM with 16 subsets. EARL range: Kaalep et al., Eur J Nucl Med Mol Imaging 2018;45:1344-61. Partial-volume effect: Soret et al., J Nucl Med 2007;48:932-45. Contouring methods: Hatt et al., AAPM TG211, Med Phys 2017;44:e1-e42. Expert outline: the autoPET lesion outline (not a GTV).

B Attenuation correction

3rd · after the break · the same simulated scan

1 · WITH AND WITHOUT AC

1 ✎ **Predict** (Colour on: bright = high) Without AC, values too high: _____

too low: _____ Right? ☐ Yes ☐ No

2 Without AC, the tumour stands out from the lung around it ☐ more ☐ the same ☐ less

3–4 Without AC, why too low in the centre, too high in lungs and skin? _____

SUV without AC? ☐ Yes ☐ No

2 · BREATHING

1 ✎ **Predict** CT with the diaphragm 2 cm higher: tumour SUVmax ☐ goes up ☐ stays ☐ goes down

Measured _____

2 Where are the errors? _____ Biggest error (table): CT _____ _____ %

3–4 What shows a mismatch in clinic? _____ How to make it smaller? _____

A SUV from the DICOM header

4th · her real images and her real DICOM header

1 · THE IMAGE

1 Find the tumour (start on the MIP). Where is it?

☐ Right upper lobe ☐ Right lower lobe ☐ Left upper lobe ☐ Left lower lobe

2 ✎ **Predict** Window 0-5 instead of 0-20: the tumour looks ☐ bigger ☐ the same ☐ smaller Right? ☐ Yes ☐ No

3 Circle on the brightest axial slice: SUVmax _____ SUVmean _____ Circle size _____ cm

4 What else lights up on the MIP? Is it cancer? _____

2 · THE DICOM HEADER

1 `SUV = C [Bq/mL] ÷ ( dose at scan start [Bq] ÷ weight [g] )`

Weight _____ kg Dose _____ MBq Injection _____ Half-life _____ min

2 Time line: injection 07:44:00 → scan starts (Series Time) 08:39:01 → this slice measured (Acquisition Time) 08:45:35.

✎ **Predict** The DICOM header has two scan times. Which do you decay the dose to? ☐ Series Time ☐ Acquisition Time

3 SUV of the hottest voxel (40713 Bq/mL) by hand.

```
fraction left = 2^( − uptake time ÷ half-life ) · dose at scan start [MBq] = 323 MBq × fraction left
SUV = 40713 Bq/mL ÷ ( dose at scan start [Bq] ÷ weight [g] ) · 1 MBq = 1000000 Bq, 1 kg = 1000 g
```

Uptake time _____ min Fraction left _____ Dose at scan start _____ MBq SUVmax _____

4 Bonus: per lean body mass (SUL). `LBM = 9270 × W ÷ (8780 + 244 × BMI)` `SUL = SUV × LBM ÷ W`

LBM _____ kg SULmax _____

3 · WHAT IF? predict the SUVmax first, then press the button

Mistake	Predict SUVmax	Page	Change
1 · Weight typed in pounds (154 instead of 70)			%
2 · Injection-room clock 30 min slow			%
3 · 10 % of the dose left in the syringe			%

4 Scanned at 90 min instead of 55. Can the calculation correct for it? ☐ Yes ☐ No Why? _____

5 Follow-up scan: what do you check in the DICOM header first? _____

Your checklist

before you trust an SUV or a PET contour · fill in at the debrief

1 · SUV calculated right? _____

2 · Does the CT match the PET? _____

3 · Reconstruction and lesion size? _____

4 · Which contouring method? _____

SUV: Boellaard et al., EANM guideline v2.0, Eur J Nucl Med Mol Imaging 2015;42:328-54. Lean body mass: Janmahasatian et al., Clin Pharmacokinet 2005;44:1051-65.
Breathing: Erdi et al., J Nucl Med 2004;45:1287-92.